



Batch: A-3 Roll No.: 16010324044
Experiment / assignment / tutorial No-04
Grade: AA / AB / BB / BC / CC / CD / DD

Signature of the Staff in-charge with date

TITLE: Two Transistor Constant Current Source

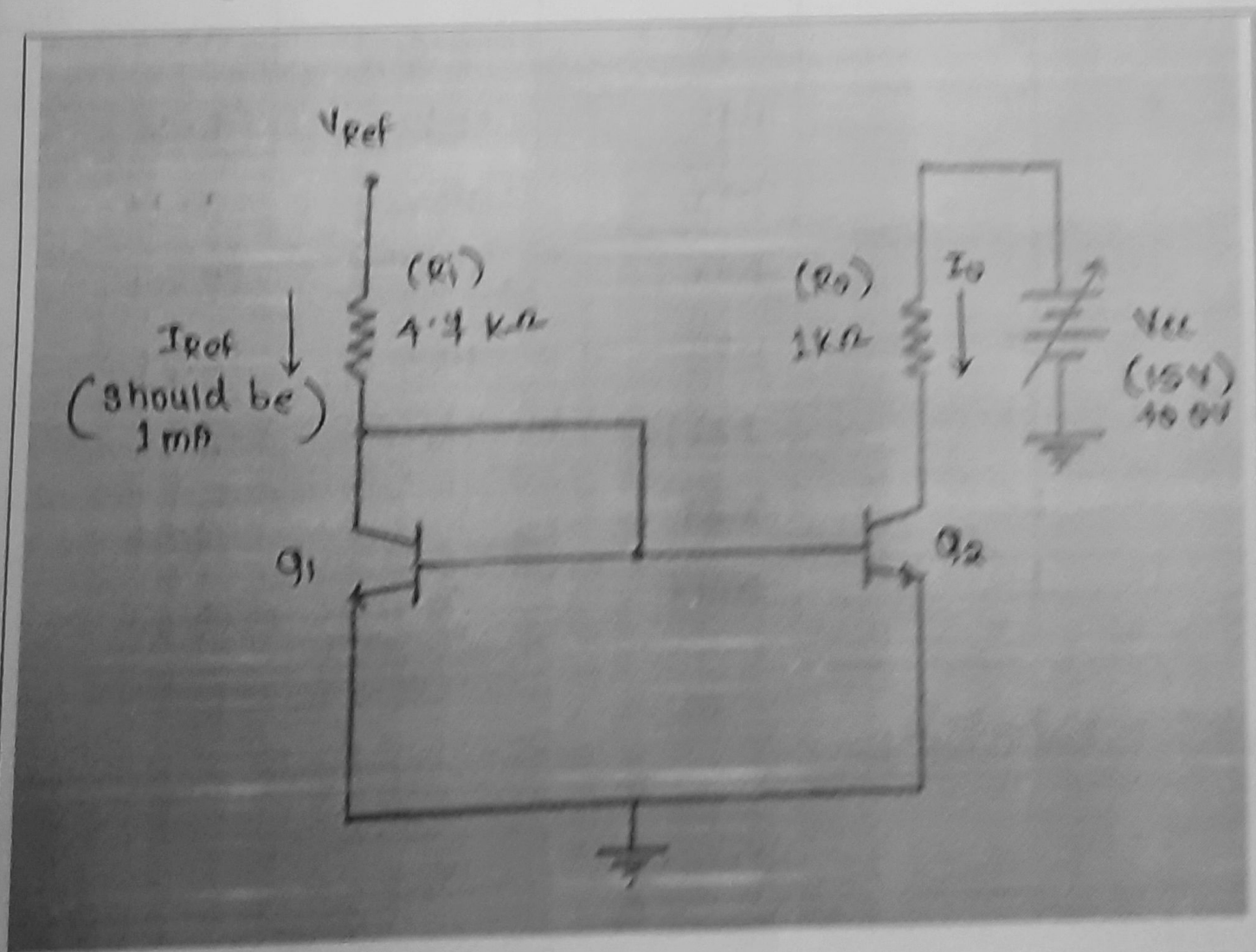
AIM: To implement basic two transistor current mirror circuit

OUTCOME: Students will be able to understand the basics of analog Integrated Circuits

PROCEDURE:

1. Assemble the circuit as shown in diagram .
2. Adjust V_{ee} such that $I_{ref} = 1 \text{ mA}$.
3. Find the values of I_o and V_c (collector voltage) of Q_2 as V_s is varied from 15V to 0V as per the observation table I.
4. Find the dynamic output resistance by plotting I_o versus V_c curve.
5. Repeat above procedure in workspace of Ltspice.

Circuit Diagram:





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Observation Table I

$V_{ref} = 5.2 \text{ V}$

$R_o = 1 \text{ k}\Omega$

Sr. No	Supply Voltage (V_s) (V)	Voltage across R_o (V_o) (across $1 \text{ k}\Omega$)	Output Current (I_o) (mA)	Collector Voltage of Q2 (V_{ce})
1.	15	1.38	1.38	13.85
2.	14	1.33	1.33	12.75
3.	13	1.30	1.30	11.77
4.	12	1.28	1.28	10.77
5.	11	1.26	1.26	9.85
6.	10	1.21	1.21	8.87
7.	9	1.18	1.18	7.95
8.	8	1.15	1.15	6.89
9.	7	1.12	1.12	5.95
10.	6	1.09	1.09	4.97
11.	5	1.06	1.06	3.97
12.	4	1.03	1.03	2.81
13.	3	1.01	1.01	2.02
14.	2	0.99	0.99	1.01
15.	1	0.93	0.93	0.13

Observation Table II

	Practically (simulation)	Theoretically
Output Resistance (r_o)	66.6 $\text{k}\Omega$	31.25 $\text{k}\Omega$



Mathematical Analysis and Theoretical Calculations

$$I_o = \frac{I_{ref}}{1 + \frac{2}{\beta}}$$

$$\beta_1 = 342, \quad \beta_2 = 336$$

$$(\beta)_{avg} = \frac{342 + 336}{2} = \underline{\underline{339}}$$

$$\text{Slope} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$R_o = \frac{1}{\text{Slope}}$$

$$V_{ref} = 5.2V$$

$$R_i = 4.7k\Omega$$

$$I_{ref} = 1mA$$

$$\left(I_{ref} = \frac{V_{ref}}{R_i} \right)$$

$$\therefore I_o = \frac{V_o}{R_o}$$

$$\text{at } V_o = 1.38V$$

$$I_o = \frac{1.38}{1k} = 1.38mA$$

$$\text{at } V_o = 1.12V$$

$$I_o = \frac{1.12}{1k} = 1.12mA$$

$$\text{at } V_o = 1.28V$$

$$I_o = \frac{1.28}{1k} = 1.28mA$$

Dynamic Resistance From Observation

$$\text{Slope}(m) = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{(1.38 - 1.06) \text{ mA}}{1.5 - 5 \text{ V}} = 0.032 (10^{-3})$$

Hence,

$$r_o = \frac{1}{\text{Slope}} = \frac{1}{m} = \frac{1}{0.032 (10^{-3})}$$

$$r_o = 31.25 \text{ k}\Omega$$

Conclusion:

This experiment successfully demonstrates the operation of a two-transistor constant current source using practical implementation.

We ~~also~~ observe that the current (I_o) (Output) is not perfectly constant as the voltage increases from 0V to 15V due to Early Effect.

Signature of faculty in-charge

Dhanya M Shah

SCALE :- x-axis $\rightarrow 1\text{ cm} = 1\text{ V}$

y-axis $\rightarrow 1\text{ cm} = 0.1\text{ mA}$

